Dual Geometry of Wavelength Space and Frequency Space
Series Handout, Papers 1–12

From three assumptions: how much of what looks quantum and spacetime-like follows *as theorems*?

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The Three Assumptions (entire inventory)

1. Reciprocal duality $\nu\lambda = 1$
2. Zero point $1/2$
3. Asymmetric one bit the conservation law $\sum \nu^2$ rides only one side of the duality

Operating principles: configuration reading and the record principle. Continuous spacetime, amplitudes, wave functions, and collapse are **not assumed**.

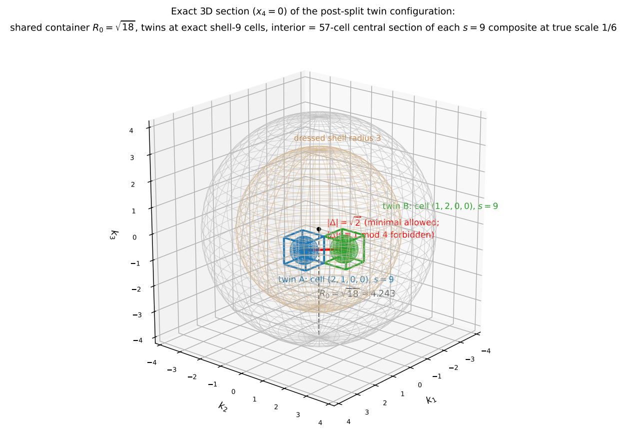
Discipline of Method

An observational model: no identification with physical quantities. Every claim is backed by exhaustive enumeration, exact identities, or machine-precision numerics. Relations to standard theory are stated as *structural correspondences*; every paper carries an explicit "claims / non-claims" section.

Foundations: Papers 1–4 (published)

Paper 1 Reciprocal duality, log representation, the 4D lattice count $N_0(R)$ `10.5281/zenodo.20588036`
Paper 2 Exact tables for $R = 0.5$–10^4 ($N_0 = 1, 9, 137, \ldots$) `10.5281/zenodo.20588038`
Paper 3 Supplement: connection to radial projection `10.5281/zenodo.20589261`
Paper 4 Splitting representation, volume gap, duality breaking, self-similar hierarchy `10.5281/zenodo.20638962`

The Twin Minimal Space Model (Paper 12)



Exact 3D section ($x_4 = 0$) of the post-split twin configuration. Inside the shared container $R_0 = \sqrt{18}$ the twins occupy actual shell-9 cells; the minimal allowed separation is $|\Delta| = \sqrt{2}$ ($|\Delta|^2 \equiv 1 \bmod 4$ forbidden). Each cell interior shows the 57-cube central section of the next-level 137-cell composite at true scale 1/6 — every position, separation, and scale is a computed value.

The Chain of Theorems (Papers 5→12)

5 Dictionary theorem: lattice counting = real standing-wave counting with zero point (exact). R^2 quantized to odd integers; the 137/105 issue resolved

6 Freezing theorem: the asymmetric bit *is* the existence condition for time-like structure. Record theorem, null structure, invariant area

7 Exclusion statistics derived: stable species $\{1, 3, 5\}$, threshold $s = 7$, unique branching 77.4% : 22.6%

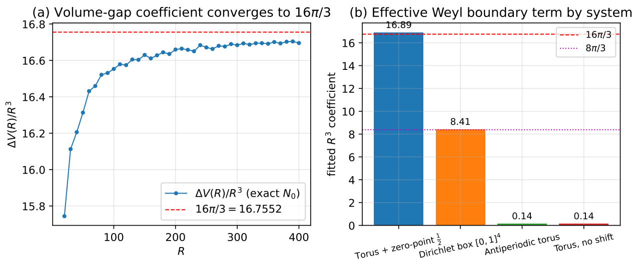
8 Accounting inverted: shared curvature → condensation; internal expansion $a \propto t^{1/2}$; vacuum as an area reservoir

9 Half-wavelength censorship: shifts $\leq \frac{1}{2}$ (saturated exactly) protect composites; existence ceiling, forced hierarchization

10 Emergent gauge: kinematically unique connection, holonomy $\pi/2$ = geodesic area, SO(4) forced

11 Dimension: $d = 4$ selected in a pincer — censorship ceiling, mod 8, self-dual seed

12 Twin model: measurement = append (absence→presence), layer resolution of entanglement, hidden axes R/Q



Verifying the dictionary (Paper 5): the volume-gap coefficient converges to $16\pi/3$ (left; exact N_0, $R \leq 400$). Right: four boundary systems — the zero-point shift generates an effective Weyl boundary of **exactly twice** the Dirichlet unit box.

The core sequence $N_0(R) = 1,\ 9,\ 137,\ 473,\ 1545,\ \ldots$
Condition: $\sum_{i=1}^{4}(|k_i| + \frac{1}{2})^2 \leq R^2$
Zero point $\frac{1}{2}$ → cell = standing wave → non-overlap = orthogonality → odd-rule coherence → exception-free splitting → quantization of R^2 — a chain with no added assumptions.

The Development: Papers 5–12 (v0.2, released 2026-06-11)

5 The Cell–Standing-Wave Dictionary	dictionary theorem, zero-point quantum, $16\pi/3$ as Weyl deficit, classification theorem	`10.5281/zenodo.20640454`
6 Conservation, Records, Time-like Structure	freezing theorem, B_4 gauge, 1+3 polar split, record theorem	`10.5281/zenodo.20640456`
7 Configuration Statistics & Relational Readout	exclusion statistics, stable species, branching ratios, holographic readout	`10.5281/zenodo.20640458`
8 Two Accountings	optimality of condensation, Jeans-type threshold, $a \propto t^{1/2}$, area law	`10.5281/zenodo.20640460`
9 Logic Waves & Half-Wavelength Censorship	amplitude-free coherence, kinematic stabilization, three-band structure	`10.5281/zenodo.20640462`
10 Emergent Gauge Structure	holonomy $\pi/2$, SO(4) forced, spin lift, $\beta = 0$	`10.5281/zenodo.20640464`
11 The Necessity of Dimension	both-endpoint theorem $[0, 4]$, mod 8, four-legs theorem, self-dual seed	`10.5281/zenodo.20640466`
12 The Twin Minimal Space Model	measurement as append, measure bifurcation, no-signaling, licensed virtual displacements	`10.5281/zenodo.20640468`

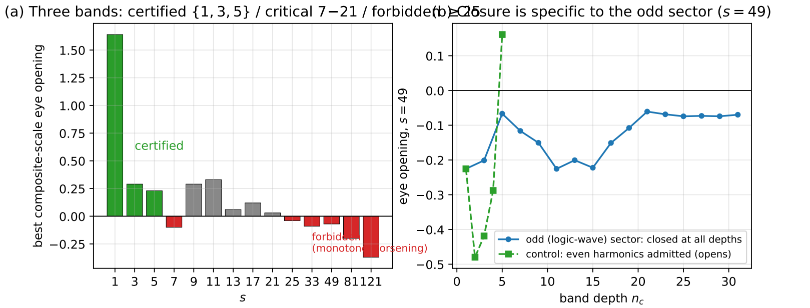
Highlights

Two-tier agreement: stable species $\{1, 3, 5\}$ and threshold $s = 7$ reached by **independent computations** — counting (Paper 7) and waves (Paper 9)

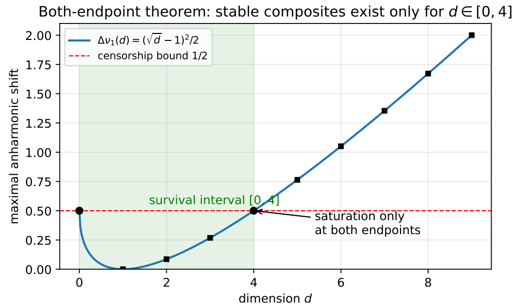
Fewer postulates: exclusion, collapse, the arrow of time, and the integer lattice move from assumptions to theorems / resolution structure

Falsifiability: amplitudes cannot couple to curvature (Paper 9) — BMV-type gravitationally mediated entanglement experiments are a live referee

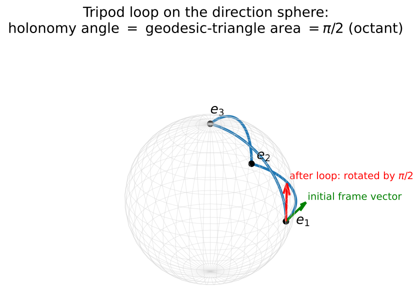
Not re-deriving is not a defect: the transport principle (Paper 12). The difference lies in candidate answers to questions standard theory leaves as postulates (collapse, the cut, time, dimension)



Three bands of existence (Paper 9): self-certification margins split into certified $\{1, 3, 5\}$ / critical / forbidden ($s \geq 25$); for $s \geq 49$ the whole odd sector closes (right). Admitting even harmonics reopens it — closure is specific to logic waves.



Both-endpoint theorem (Paper 11): $\Delta\nu_1(d) = (\sqrt{d} - 1)^2/2 \leq \frac{1}{2} \iff d \in [0, 4]$, saturated only at the endpoints — $d = 4$ is critical at the ceiling.



The first holonomy (Paper 10): transport around the tripod loop rotates by $\pi/2$ — exactly the area of the geodesic octant. Discrete transport reproduces continuous curvature.

Research notes (Supplements 1–41), all verification scripts, and figure-generation code: `github.com/WurabeSeiji/ai-chat-logs-open` (folder "Dual Geometry of Wavelength Space and Frequency Space"). All figures are exact computations (no schematic diagrams). This series is an observational model; it does not claim to derive or modify physical theory.